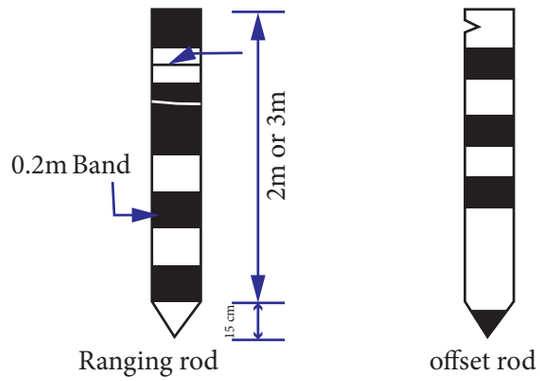
4. Pegs

Pegs are made of wood. These are 25 mm square and 150 mm long, tapered at one end. It is used to mark the survey stations by driving it into the ground. Iron or steel pegs of 12 mm diameter can also be used as pegs.

5. Ranging Rods

Ranging rods are made of well seasoned, straight grained timber or steel tubular rods.

These are circular (or) octagonal in cross section of 25 mm diameter and 2 m or 3 m long. The bottom end is tapered and fitted with a metal shoe to insert it into the ground. They are painted with alternate bands of red and white (0.2 m band length). A flag can also be attached to the top of the ranging rod.



Ranging rod are used to locate the survey stations and to range the survey lines.

6. Offset rods

It is similar to ranging rod. It is provided with a recessed hook for pulling the chain through any obstruction.

It can be used for locating stations and ranging the lines.

7. Plumb-bob

It is made of steel in conical shape with a thread connected at the centre. Generally it is used for centering a station point.

While measuring distances on the sloping ground, it is used to transfer the point to the ground by suspending the plumb bob.



Plumb-bob

8. Cross Staff

Cross staff is used for

i. Establishing perpendicular offset from a given point to a line

ii. Setting out a right angle at a given point on a line

The two forms of cross staff commonly used are:

a. Open cross staff

b. French cross staff

a. Open cross staff

It consists of two pairs of vertical slits providing two lines of sight mutually at right angles. Each pair consists of two vanes, one is eye vane and the other is objective vane.

b. French cross staff

It consists of an octagonal brass tube with slits on all eight sides. It has an alternate vertical slit and an opposite vertical window, with a vertical cross hair

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